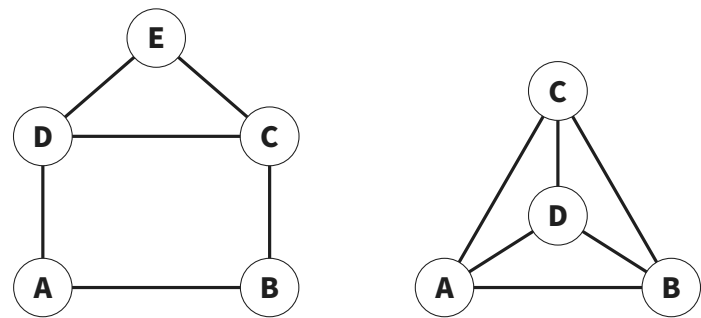


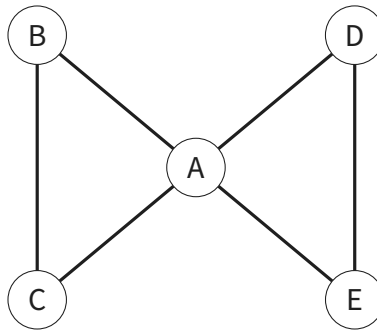
Problem 1: Try to use every line once without jumping. Revisiting a dot is allowed. Mark used lines with counters. Try the house, then the triangle with center D. Choose two different starting dots on each. Record a complete route, or your attempted route and unused lines.



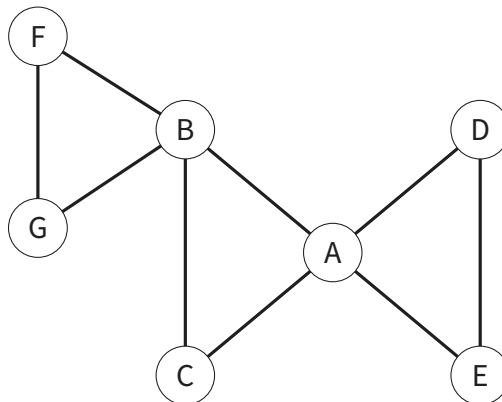
Map	Start	Route	Unused lines
House			
House			
Triangle			
Triangle			

Problem 2: Count each dot’s incident lines, called its degree. Circle the odd-degree dots. Choose your complete house route and pair the arrival/departure lines at each internal visit. Record the dots with one unpaired line.

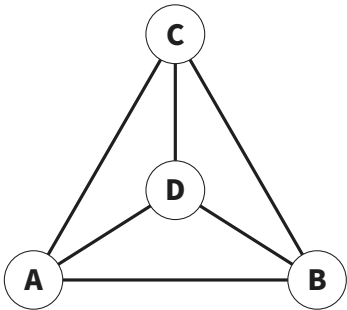
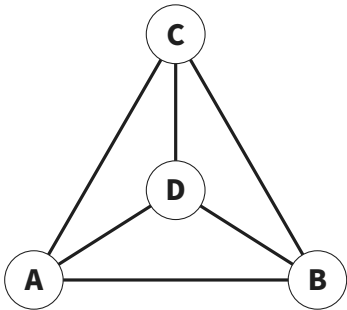
Problem 3: Walk A–B–C–A and mark those roads used. Walk the remaining loop A–D–E–A. Write each loop on a paper strip. Join the strips at A to make one route using all six roads. Record and trace the joined route.



Problem 4: The map below adds a loop at B. Start with your route from Problem 3, then insert B–F–G–B at a visit to B. Record the joined route and check every road is used once. Count the degree at B before and after the new loop.



Problem 5: Now travel every road at least once and return to A. Roads may be repeated; each traversal costs 1 step. Find a route on the left map and record its visited dots. Try a different route on the right map with fewer steps, or match your best count. Mark each repeated road with an extra pencil line.



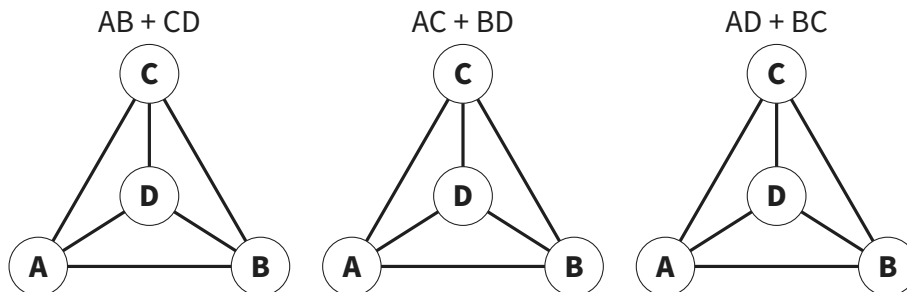
Trial	Route	Steps
1		
2		

Problem 6: Count how many times your best route uses each road. Record the original six crossings separately from the repeats. At each dot, count all arrival and departure crossings, including repetitions. Circle any odd total.

Road	AB	AC	AD	BC	BD	CD
Times used						

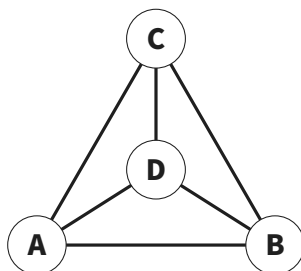
Problem 7: Add one extra copy of a road to a clean map. Count how many odd degrees change to even. Then add one more road copy that makes all degrees even. Record the two roads.

Problem 8: Pair the four odd dots in each way shown. Draw a second copy of each named road on its map. Then find a closed route using every original and added road once. Record its length.

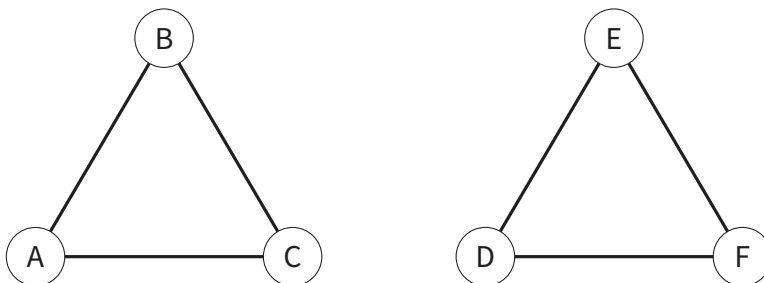


Pairing	Closed route	Length
AB + CD		
AC + BD		
AD + BC		

Problem 9: Now the truck may finish at a different dot. Add just one road copy to the original map below, then find a route using every original and added road once. Record its endpoints and length.



Problem 10: Add three roads between these two triangles so the map is connected and all six dots have odd degree. Curve roads around other dots; a crossing without a dot is not a junction. Count the degrees. Then choose three roads to repeat so every degree becomes even. Record a closed delivery route and its length.



Problem 11: This map also has six odd dots. Start at A, visit every road, and return to A. Mark each repeated road and record your route. Compare how many extra crossings the two maps need.

